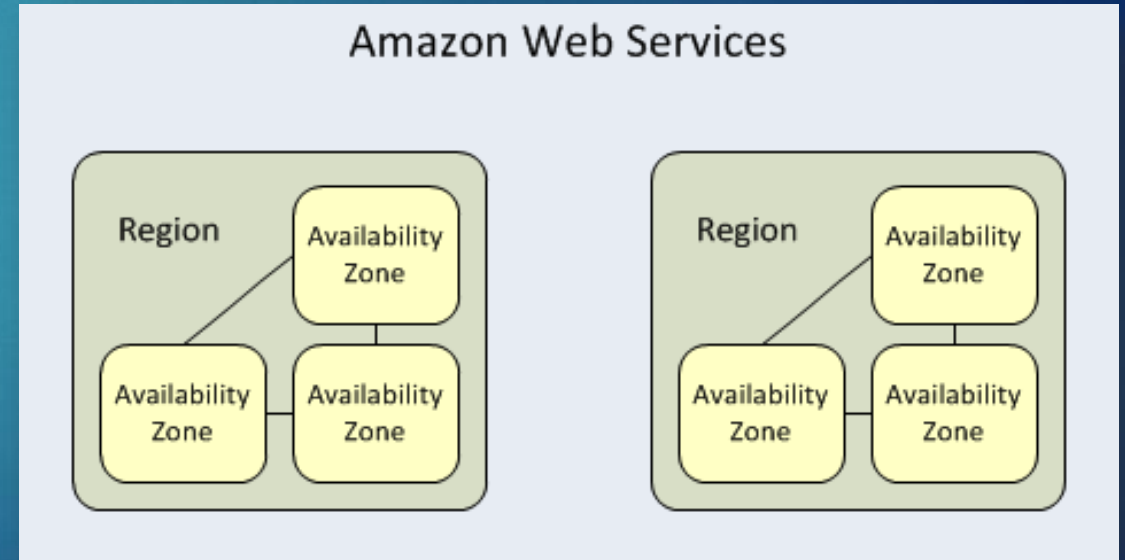


- ▶ S3 buckets are tied to a specific AWS Region
 - ▶ True ☒
 - ▶ False
- ▶ Which AWS storage option is the most cost-effective for storing large amounts of static data that doesn't require frequent access?
 - ▶ EBS
 - ▶ EFS
 - ▶ S3 ☒
 - ▶ Instance Store
- ▶ An EBS volume persists independently from the running life of an EC2 instance
 - ▶ True ☒
 - ▶ False

- ▶ A Region contains 2 more or Availability Zones
- ▶ Regions
 - ▶ Separate geographic area where your EC2 instances are running
 - ▶ Each is designed to be completely isolated from the other to provide fault tolerance and stability
- ▶ Availability Zones
 - ▶ Isolated, but other zones in same region are connected



- ▶ Storage models
 - ▶ Simple Storage Service (S3)
 - ▶ Cheapest for data storage and can be accessed from anywhere via a URL
 - ▶ Elastic Block Store (EBS)
 - ▶ Mount to a single EC2 instance
 - ▶ Only available in a particular region, but faster than S3 and EFS
 - ▶ Elastic File System (EFS)
 - ▶ Available on multiple regions and can be accessed by up to *thousands* of EC2 instances at once
 - ▶ Best for large very large quantities of data such as large analytic workloads
 - ▶ Other specialized models:
 - ▶ Glacier, Storage Gateway, Snowball, Snowball Edge and Snowmobile



- ▶ For your S3 website, how much do you think it costs to host per month?
 1. \$1-\$3/month
 2. \$5-\$10/month
 3. \$10-\$20/month
 4. > \$20/month
- ▶ Answer
 - ▶ Typically, it will cost \$1-3/month if you are outside the AWS Free Tier limits
 - ▶ If you are eligible for AWS Free Tier and within these limits, hosting your static website will cost around \$0.50/month

AWS Advanced

Engineering Cloud Software Systems

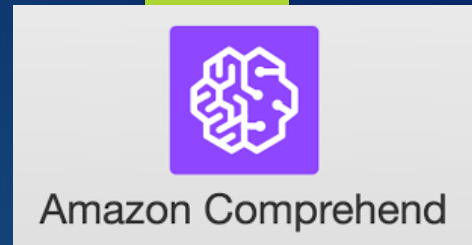
Department of Software Engineering
Rochester Institute of Technology



Homework Assignment and Team Project

- ▶ You will research AWS tutorials of your choice and these will later be shared with the class
- ▶ The purpose of the tutorials is to:
 - ▶ Get you and your team familiar with AWS services faster to help spark ideas for the team project
 - ▶ Share your research with the class to help them better understand technologies they may not have looked at
- ▶ We will explore several AWS services in future classes but let's look at a few that won't be covered in class that you may want to research further for your project
 - ▶ Several teams used these services on past projects

AWS Comprehend



- ▶ A natural-language processing (NLP) service that uses machine learning to uncover information in unstructured data

Input

Entities

Key phrases

Language

PII

Sentiment

Syntax

Analyzed text

Hello Zhang Wei. Your AnyCompany Financial Services, LLC credit card account 1111-0000-1111-0000 has a minimum payment of \$24.53 that is due by July 31st. Based on your autopay settings, we will withdraw your payment on the due date from your bank account XXXXXX1111 with the routing number XXXXX0000.

Your latest statement was mailed to 100 Main Street, Anytown, WA 98121.

After your payment is received, you will receive a confirmation text message at 206-555-0100.

If you have questions about your bill, AnyCompany Customer Service is available by phone at 206-555-0199 or email at support@anycompany.com.



Output

Entity	Type	Confidence
Zhang Wei	Person	0.99+
AnyCompany Financial Services, LLC	Organization	0.99+
1111-0000-1111-0000	Other	0.87
\$24.53	Quantity	0.99+
July 31st	Date	0.99+
XXXXXX1111	Other	0.85
XXXXX0000	Other	0.83
100 Main Street, Anytown, WA 98121	Location	0.99+
206-555-0100	Other	0.99+
AnyCompany	Organization	0.97

- ▶ Project Ideas:

- ▶ Process a data feed to find out what people are saying about a certain entity (person, topic, etc.)
- ▶ Analyze other content feeds (e.g. News) to look for relationships between common entities

AWS Rekognition



- Identify objects, people, text, scenes, and activities in images and videos, as well as detect any inappropriate content

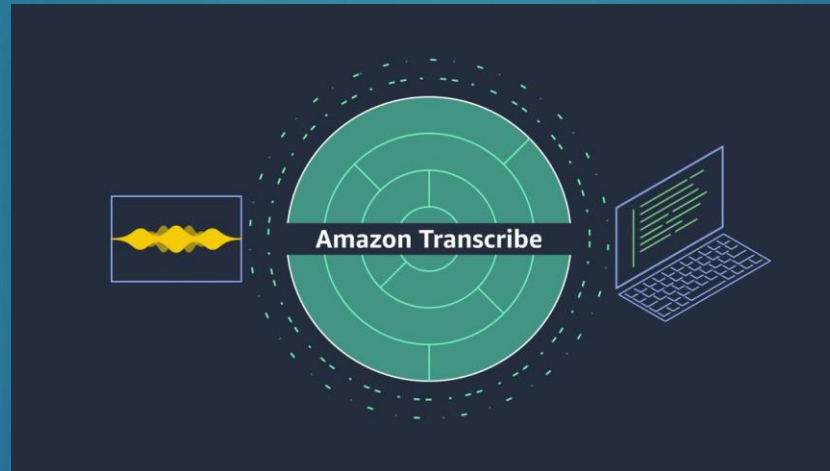


- Project Ideas
 - Process a live or recorded image feed to determine certain characteristics:
 - How many people are in the image?
 - What other things are in the image (e.g. animal)?



AWS Transcribe

- ▶ Uses a deep learning process called automatic speech recognition (ASR) to convert speech to text quickly and accurately
- ▶ It is designed to process audio input from a variety of sources such as microphones, audio or video files and provide high quality transcriptions for search and analysis

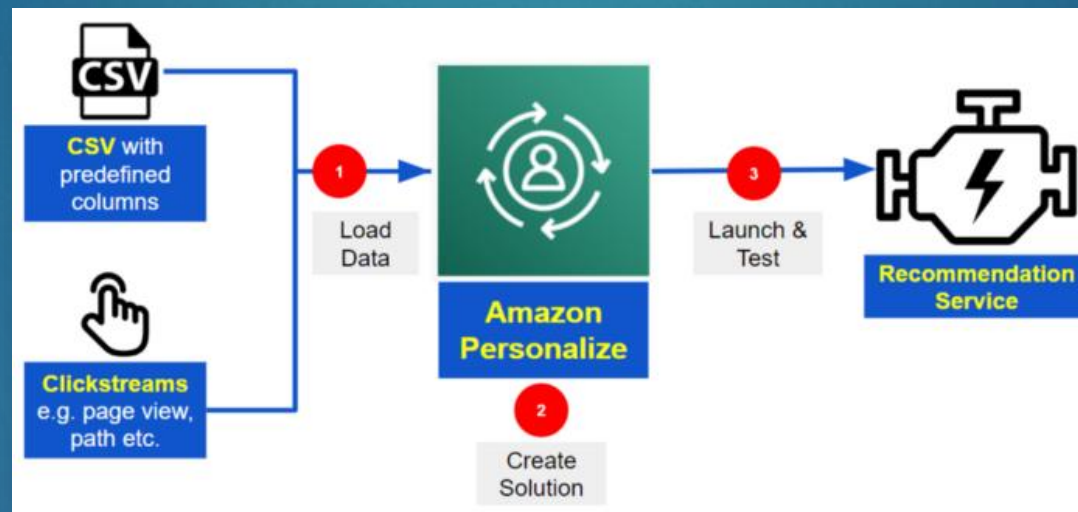


- ▶ Project Ideas:
 - ▶ Process recorded audio feed (e.g. lecture) and use the output to create a summarization
 - ▶ Process a live audio feed and search for key entities using AWS Comprehend



AWS Personalize

- ▶ Enables developers to build applications with the same machine learning (ML) technology used by Amazon.com for real-time personalized recommendations – no ML expertise required

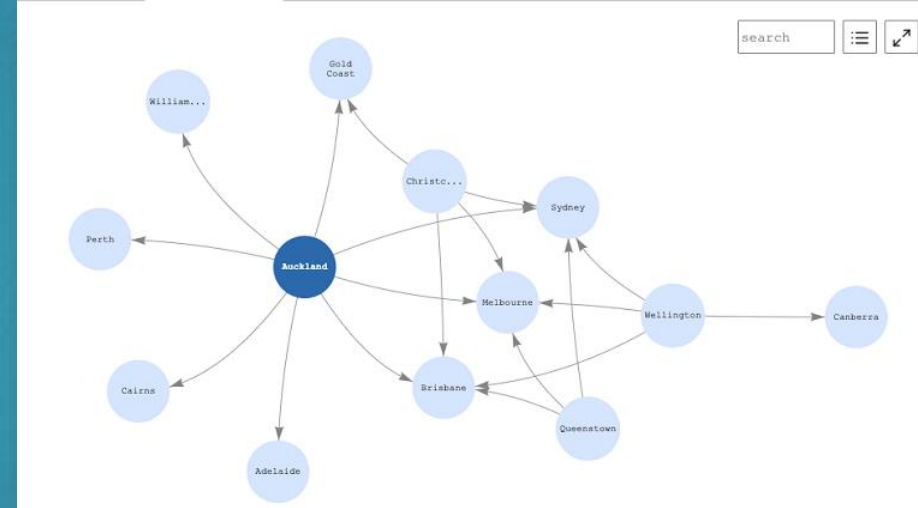
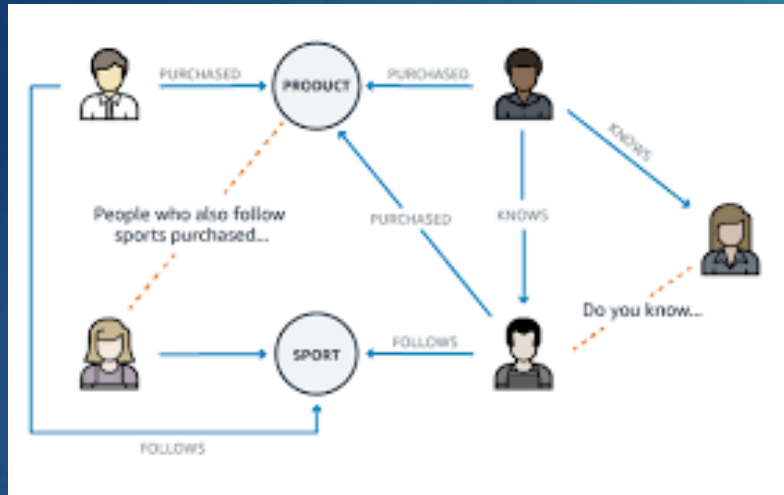


- ▶ Project Ideas:
 - ▶ Create your own specialized recommendations web site (eStore, travel, restaurant, etc.)
 - ▶ Process a resume and recommend relevant jobs using a publicly available Jobs API

AWS Neptune



- Fully managed graph database service that makes it easy to build and run applications that work with highly connected datasets



- Project Ideas:
 - Process a data feed and extract common entities (via AWS Comprehend) to visually display a graph
 - Using a data feed of locations of airports, determine the possible routes from one city to another

Other Technologies to Explore

▶ **AWS Amplify**

- ▶ A set of tools and services that can be used together or on their own, to help front-end web and mobile developers build scalable full stack applications
- ▶ This is not permitted for any class projects, but feel free to learn about this

▶ **Amazon Polly**

- ▶ Service that turns text into lifelike speech, allowing you to create applications that talk, and build entirely new categories of speech-enabled products

▶ **Amazon Translate**

- ▶ A neural machine translation service that delivers fast, high-quality, affordable, and customizable language translation

▶ **Amazon Lex**

- ▶ A service for building conversational interfaces into any application using voice and text

Need Data?

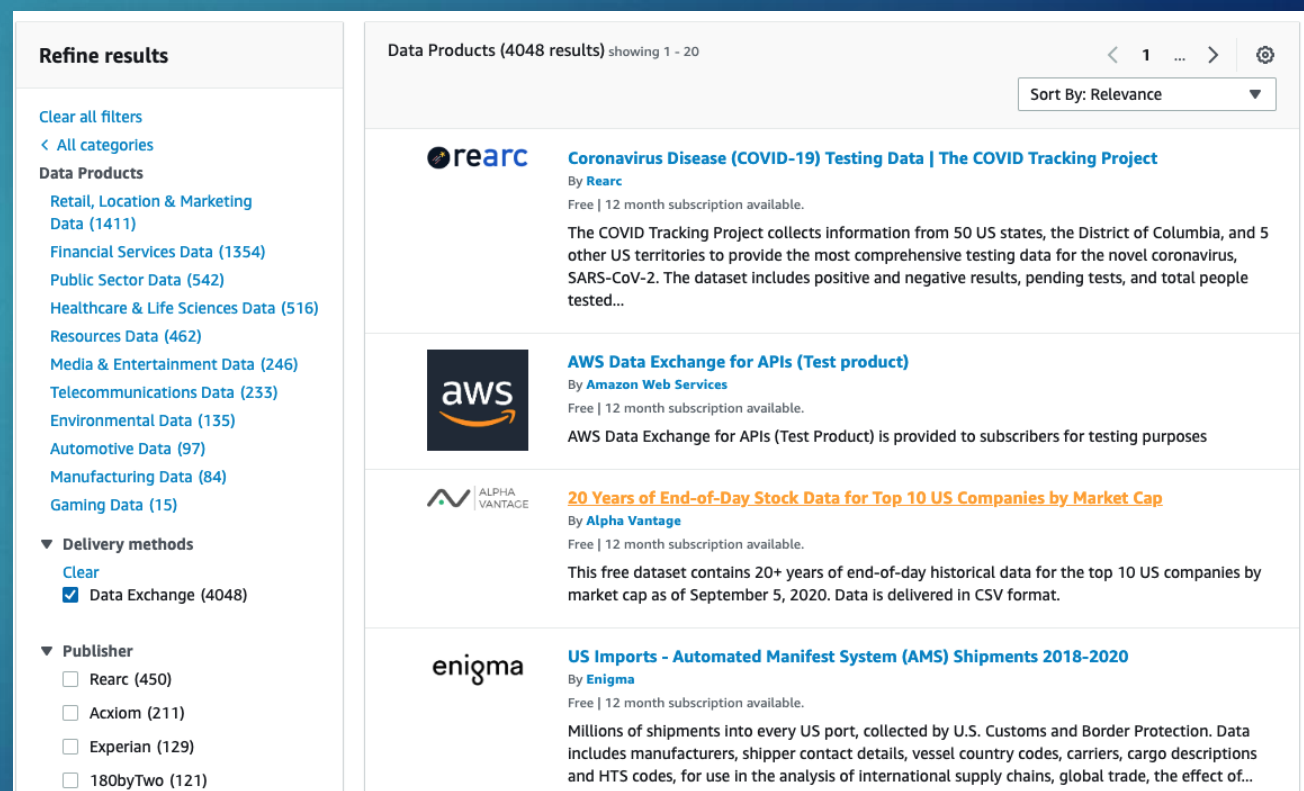
- Check out: <https://aws.amazon.com/data-exchange/>



AWS Data Exchange
Easily find, subscribe to, and use third-party data in the cloud

Browse 3,500+ third-party data sets

- Extensive data set catalog
- Better data technology with AWS integration
- Streamlined data procurement and governance
- Easy to use for data files, tables, and APIs



Refine results

Clear all filters
< All categories

Data Products

- Retail, Location & Marketing Data (1411)
- Financial Services Data (1354)
- Public Sector Data (542)
- Healthcare & Life Sciences Data (516)
- Resources Data (462)
- Media & Entertainment Data (246)
- Telecommunications Data (233)
- Environmental Data (135)
- Automotive Data (97)
- Manufacturing Data (84)
- Gaming Data (15)

▼ **Delivery methods**

Clear

☒ Data Exchange (4048)

▼ **Publisher**

- ☐ Rearc (450)
- ☐ Acxiom (211)
- ☐ Experian (129)
- ☐ 180byTwo (121)

Data Products (4048 results) showing 1 - 20

Sort By: Relevance

- rearc** **Coronavirus Disease (COVID-19) Testing Data | The COVID Tracking Project**
By **Rearc**
Free | 12 month subscription available.
The COVID Tracking Project collects information from 50 US states, the District of Columbia, and 5 other US territories to provide the most comprehensive testing data for the novel coronavirus, SARS-CoV-2. The dataset includes positive and negative results, pending tests, and total people tested...
- aws** **AWS Data Exchange for APIs (Test product)**
By **Amazon Web Services**
Free | 12 month subscription available.
AWS Data Exchange for APIs (Test Product) is provided to subscribers for testing purposes
- ALPHA VANTAGE** **20 Years of End-of-Day Stock Data for Top 10 US Companies by Market Cap**
By **Alpha Vantage**
Free | 12 month subscription available.
This free dataset contains 20+ years of end-of-day historical data for the top 10 US companies by market cap as of September 5, 2020. Data is delivered in CSV format.
- enigma** **US Imports - Automated Manifest System (AMS) Shipments 2018-2020**
By **Enigma**
Free | 12 month subscription available.
Millions of shipments into every US port, collected by U.S. Customs and Border Protection. Data includes manufacturers, shipper contact details, vessel country codes, carriers, cargo descriptions and HTS codes, for use in the analysis of international supply chains, global trade, the effect of...

- Outside of AWS, there are many more “free open data” sets available

Term Project

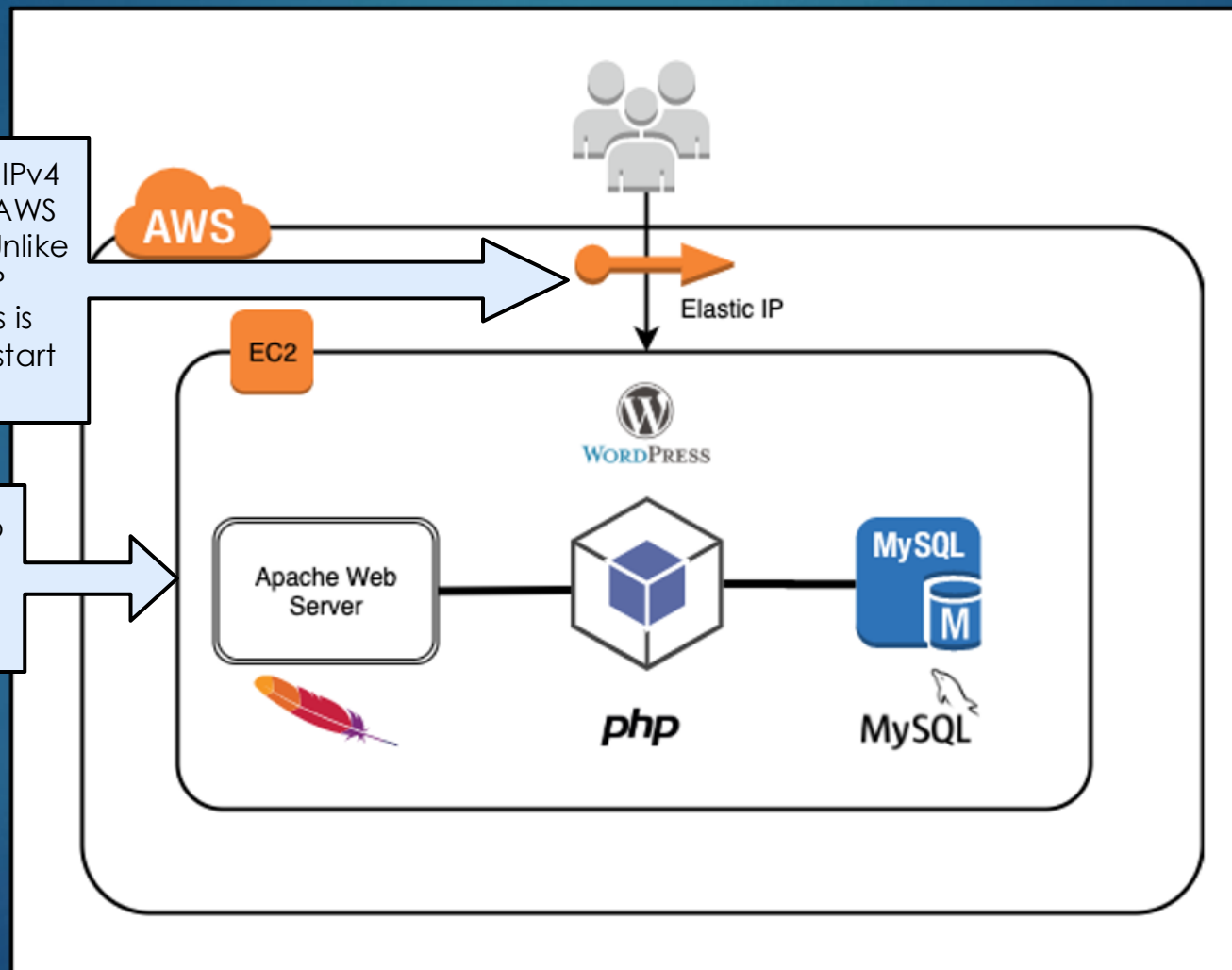
- ▶ Either choose your own or select from the pre-defined Project Templates
- ▶ Choose your own:
 - ▶ Teams will come up with their own idea on a project using AWS services. This option offers the most flexibility for teams that want to learn and explore AWS services of their choice
- ▶ Select a Project Template:
 - ▶ Teams will select 1 of the provided project templates, which includes a high-level overview description of a project including the AWS technologies that should minimally be used
- ▶ Note that all projects must be pre-approved by Instructor (check Syllabus for date)
- ▶ In addition, teams must incorporate at least one additional AWS service of their choice
- ▶ Project Details: myCourses (see Helpful Links – “Term Project”)

Last Activity – Install WordPress on EC2

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An Elastic IP address is a static IPv4 address associated with your AWS account in a specific Region. Unlike the auto-assigned public IP address, an Elastic IP address is preserved after you stop and start your instance

Whenever you start/stop an EC2 instance, AWS will auto-assign a public IP address

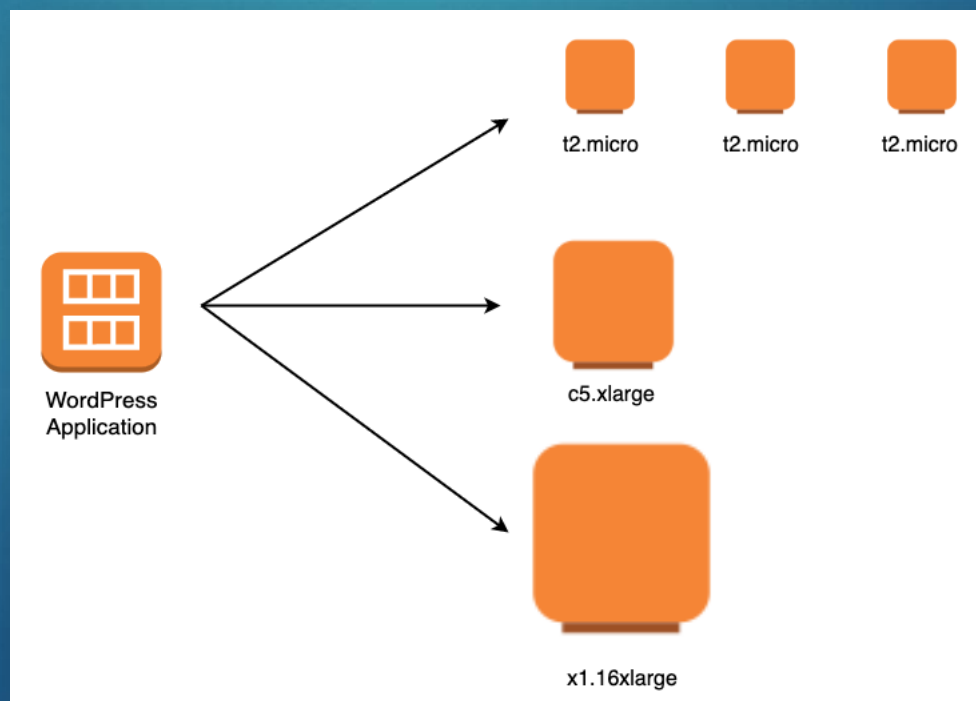


Questions:

- ▶ What if I wanted to upgrade to a larger EC2 instance?
- ▶ How can I share my EC2 instance with others?
- ▶ How could something like this be developed across a team?
- ▶ How does AWS secure my EC2 instance?
- ▶ What if I want some of my AWS services (e.g. database) to not be accessible from the Internet?

Amazon Machine Image (AMI)

- ▶ AMI is the “Golden Master”
- ▶ It provides the information required to launch an instance of any size
- ▶ You can launch multiple instances from a single AMI when you need multiple instances with the same configuration



Creating an AMI Image

- ▶ Using your completed WordPress EC2 instance, you can create an AMI image that can be reused
- ▶ To do this, select **Actions > Images and Templates > Create Image**

The screenshot displays the AWS Management Console's 'Instances' page. At the top, there's a header with 'Instances (1/1)' and an 'Info' link. Below this is a search bar labeled 'Filter instances'. A table lists the instance details:

☒	Name	Instance ID	Instance state	Instance type	Status check
☒	WordPress	i-0a13cc466f3ecd...	Running	t2.micro	Initializing

To the right of the table, the 'Actions' menu is expanded, showing options like 'Connect', 'View details', 'Manage instance state', 'Instance settings', 'Networking', 'Security', 'Image and templates', and 'Monitor and troubleshoot'. The 'Image and templates' option is selected, and a sub-menu is visible with 'Create image', 'Create template from instance', and 'Launch more like this'.

Creating an AMI Image

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#1 - Name your image and select "Create image"

Create image [Info](#)

An image (also referred to as an AMI) defines the programs and settings that are applied when you launch an EC2 instance. You can create an image from the configuration of an existing instance.

Instance ID
i-0a13cc466f3ecdd89

Image name
WordPress
Maximum 127 characters. Can't be modified after creation.

Image description - optional
WordPress application
Maximum 255 characters

No reboot
☐ Enable

No tags associated with the resource.
[Add tag](#)
You can add 50 more tags.

[Cancel](#) [Create image](#)

#2 - The image is now available under:
Images > AMIs

New EC2 Experience

Launch EC2 Image Builder Actions

Owned by me Filter by tags and attributes or search by keyword

Name	AMI Name	AMI ID	Source	Owner	Visibility	Status
	WordPress	ami-026478a127382b4e3	075837463923/...	075837463923	Private	available

Image: ami-026478a127382b4e3

Details Permissions Tags

AMI ID	ami-026478a127382b4e3	AMI Name	WordPress
Owner	075837463923	Source	075837463923/WordPress
Status	available	State Reason	-

Edit

Creating an AMI Image

- ▶ Your image is now available under “My AMIs” when you configure an EC2 instance to launch

Choose an Amazon Machine Image (AMI)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

Search for an AMI by entering a search term e.g. "Windows" ▼

Quickstart AMIs (45)
Commonly used AMIs

My AMIs (2)
Created by me

AWS Marketplace AMIs (5913)
AWS & trusted third-party AMIs

Community AMIs (500)
Published by anyone

Refine results

Clear all filters

▼ Owner

☒ Owned by me

☐ Shared with me

▼ OS category

☐ All Linux/Unix

☐ All Windows

All products (1 filtered, 2 unfiltered) < 1 >

WordPress

ami-096d07ad52e52ab5d

Platform: Other Linux Architecture: x86_64 Owner: 702897622105 Publish date: 2022-08-06 Root device type: ebs Virtualization: hvm ENA enabled: Yes

Select

EC2 Instances - Sharing

- ▶ So, if you are building an application on an EC2 instance, can you share this with your team members?
 - ▶ You cannot share a specific EC2 instance with your team

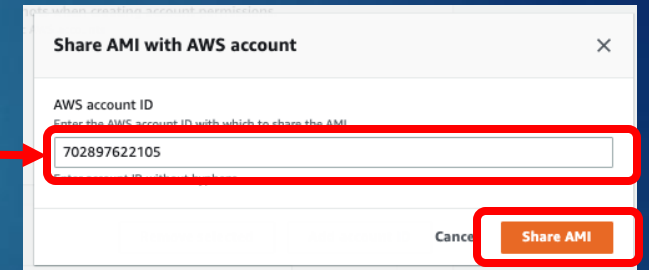
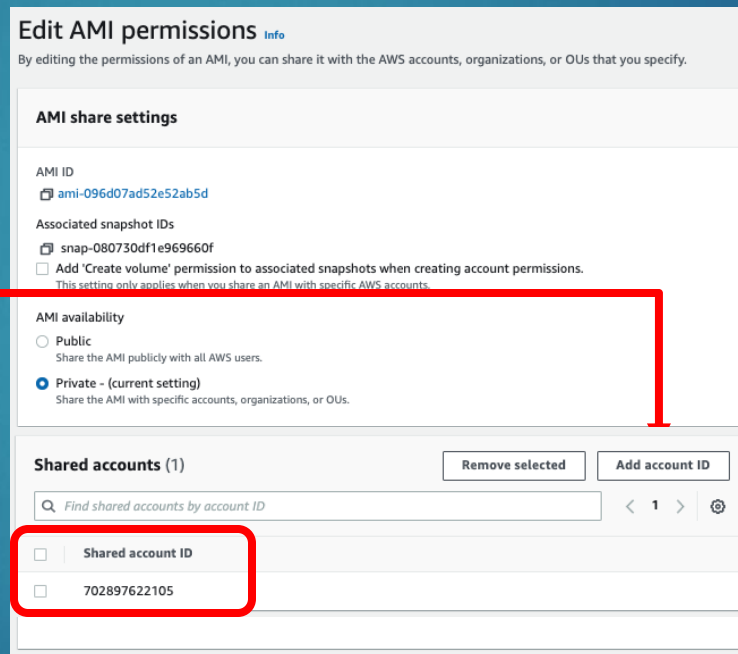
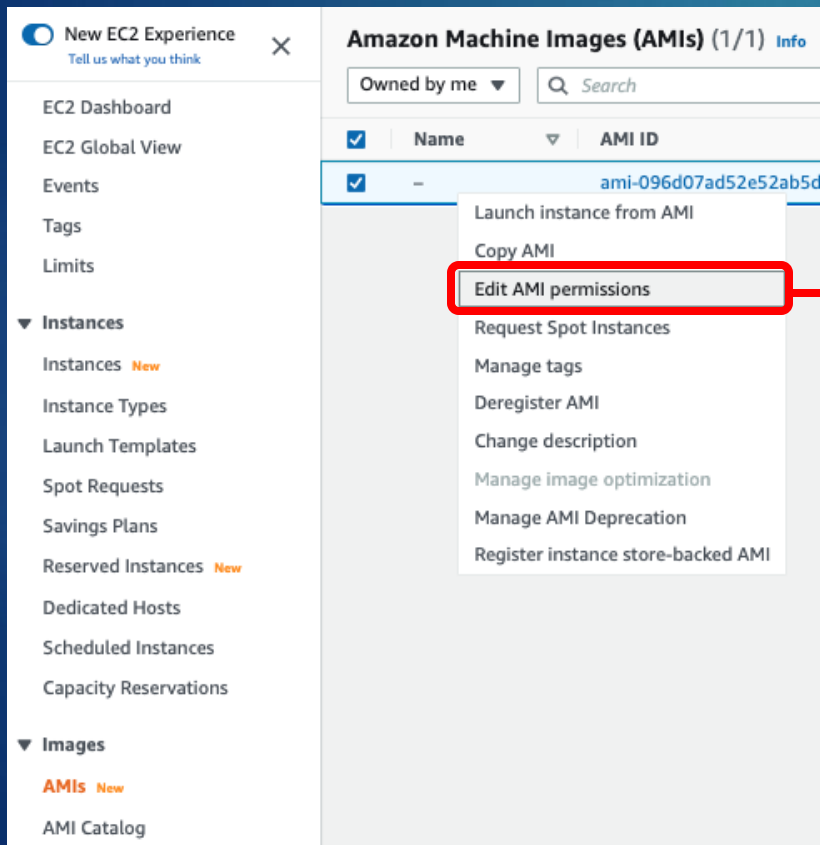


- ▶ You can share an AMI image

Sharing an AMI Image

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- ▶ Under AMI, Select AMI image and select **Edit AMI permissions**
- ▶ Click “Add account ID” and add the account # of your team members and click “Share AMI”



- ▶ The account ID will now appear under Shared accounts

AWS Account ID

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- ▶ Your account ID is in the upper right corner

The screenshot displays the AWS Management Console interface. At the top, the navigation bar includes the AWS logo, a 'Services' menu, a search bar, and the current region 'N. Virginia'. On the far right of the navigation bar, the user's identity is shown as 'voclabs/user2046505=Test_Student @ 7028-9762-2105'. A dropdown menu is open from this user identity, and the first item, 'Account ID: 7028-9762-2105', is highlighted with a red rectangular box. Below the navigation bar, the main content area is titled 'Console Home' and features a 'Recently visited' section with links to EC2, Amazon Textract, S3, CloudShell, and AWS Amplify. A 'Welcome to AWS' section is also visible on the right side of the console.

Create Instance with a Shared AMI

- ▶ Team members that had their account # added, the shared image will appear under “My AMIs” when you create an EC2 instance

Choose an Amazon Machine Image (AMI)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

Search for an AMI by entering a search term e.g. "Windows"

Quickstart AMIs (45)
Commonly used AMIs

My AMIs (2)
Created by me

AWS Marketplace AMIs (5913)
AWS & trusted third-party AMIs

Community AMIs (500)
Published by anyone

Refine results

Clear all filters

▼ Owner

☐ Owned by me

☒ Shared with me

OS category

Linux/Unix

All products (1 filtered, 2 unfiltered)

Test image - MZ

ami-077d3e095947767e5

Platform: Other Linux Architecture: x86_64 Owner: 479985889380 Publish date: 2022-07-24 Root device type: ebs ENA enabled: Yes

Select

Make sure you check the “Shared with me” checkbox

AMI and OS

- ▶ Available in Windows, Linux and Mac
 - ▶ In class, will be working exclusively in Linux
 - ▶ For your team project, you can pick whatever OS you want, but be aware they are all not free
- ▶ There are several Linux OS available
 - ▶ These include Redhat, Ubuntu and SUSE
- ▶ We have been using Amazon Linux AMI
 - ▶ Integrates easier with AWS services and tools
 - ▶ Optimized for use in Amazon EC2 with a latest and tuned Linux kernel version



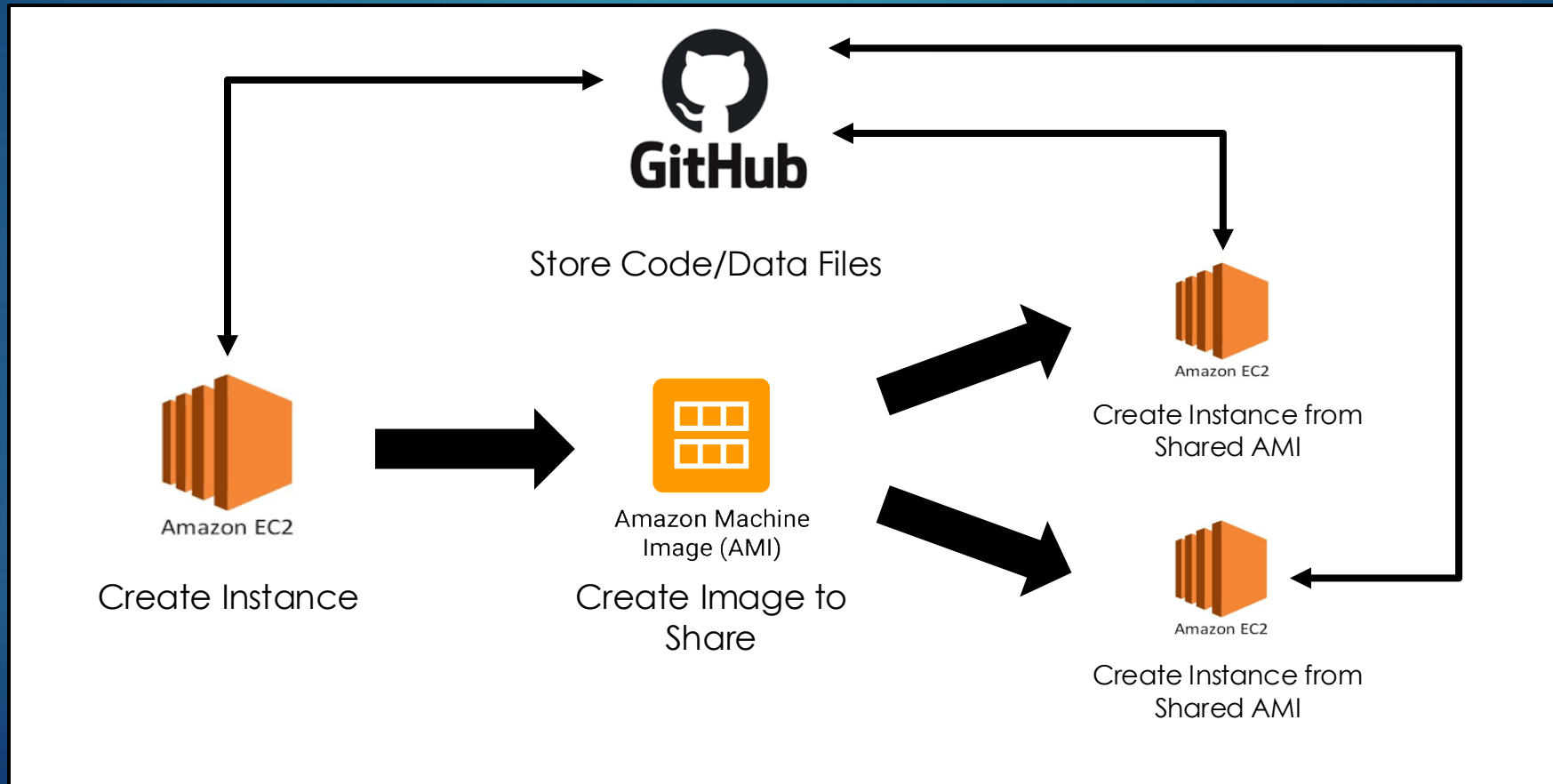
EC2 Instances – Watch your usage!

- ▶ You will get charged even if you are using it
- ▶ Only terminate if you do not plan to use your instance any more
- ▶ If you want to save your work on your EC2 instance, you can stop it and restart later
 - ▶ You will not be charged for the instance while in a stopped state
 - ▶ You will be charged other resources that are connected to it (e.g. EBS)



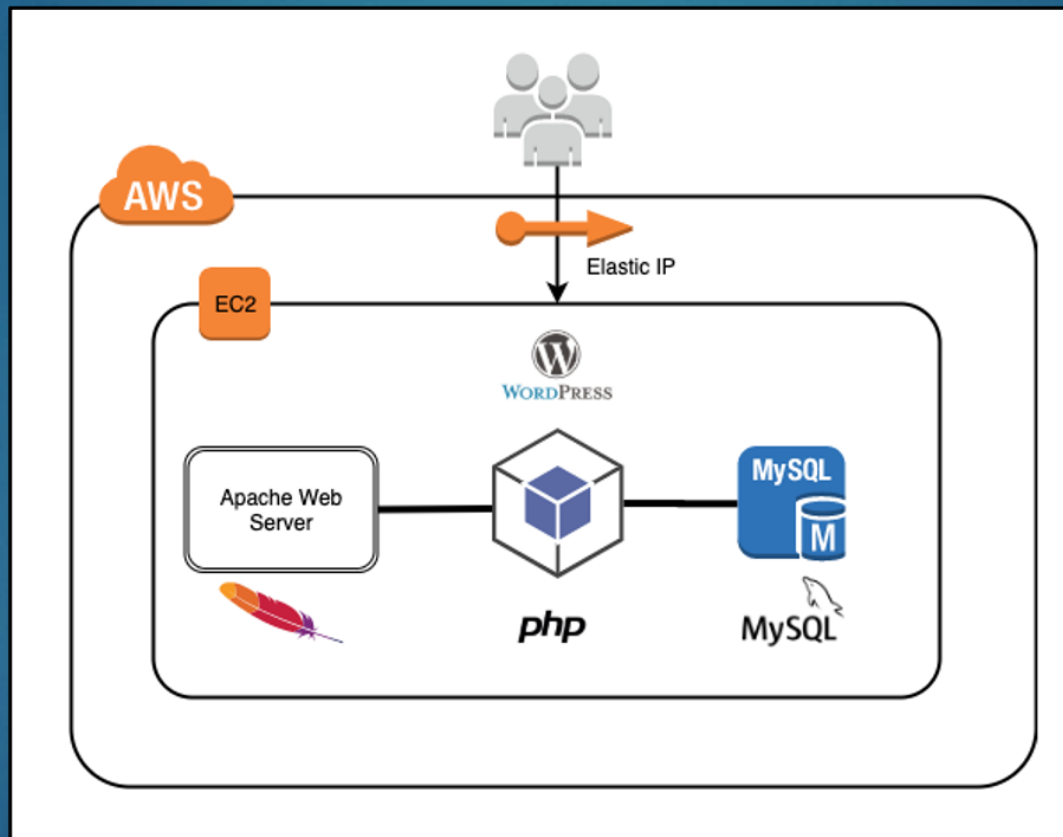
Sharing Code and Instances

- ▶ Share your EC2 instance via AMI for other team members to use
- ▶ Use GitHub to store all team's code and data



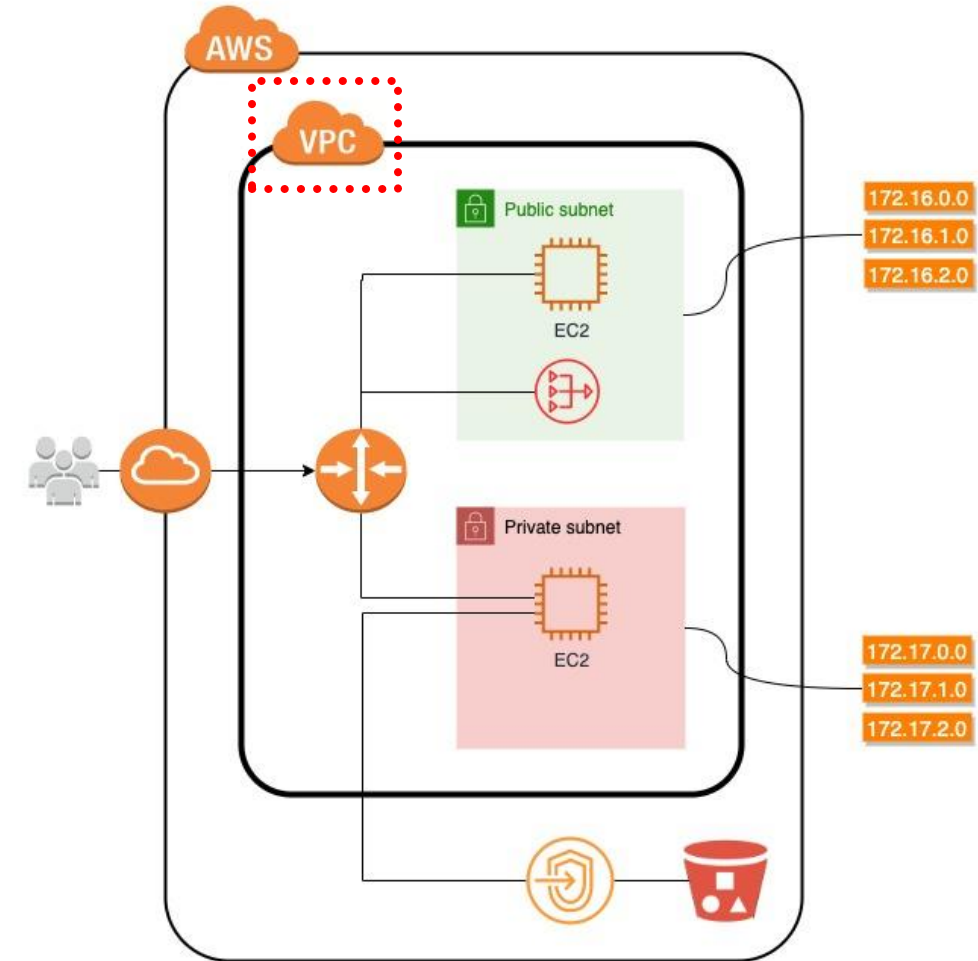
How does AWS secure my EC2 instance?

27



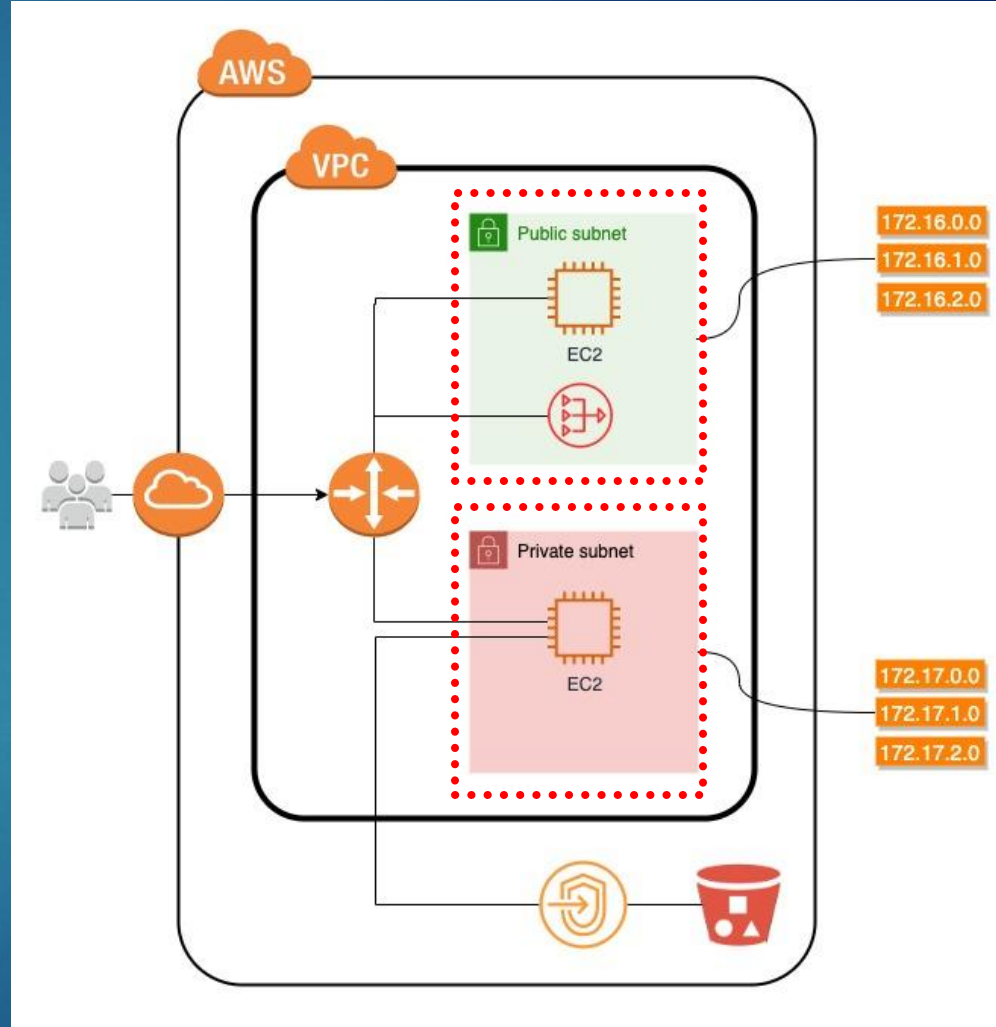
Virtual Private Cloud (VPC)

- ▶ A VPC is a virtual network dedicated to your AWS account
- ▶ Logically isolated from other virtual networks in AWS
- ▶ You can launch your AWS resources, such as Amazon EC2 instances, into your VPC
- ▶ You can specify an IP address range for the VPC, add subnets, associate security groups, and configure route tables



Subnet

- ▶ A Subnet is a range of IP addresses in your VPC
- ▶ Use a public subnet for resources that must be connected to the internet
- ▶ Resources in a private subnet can only access the internet using a network address translation (NAT) gateway that resides in the public subnet
- ▶ You can assign a subnet to a specific IP address, but it is not required
 - ▶ If you don't, AWS will automatically address it from the IP address range you assigned to it
- ▶ There is a limit of 200 subnets that can be created per VPC



Calculating IP addresses for VPC vs. Subnets

- ▶ When you create a VPC, you must specify an IPv4 CIDR (Classless Inter-Domain Routing) block for the VPC. Examples:
 - ▶ $10.0.0.0/24 \rightarrow 2^{32} - 2^{24} = 2^8 = \mathbf{256}$ IP addresses for your VPC (10.0.0.0 – 10.0.0.255)
 - ▶ $10.0.0.0/20 \rightarrow 2^{32} - 2^{20} = 2^{12} = \mathbf{4096}$ IP addresses for your VPC (10.0.0.0 – 10.0.15.255)
 - ▶ The allowed block size is between a **/16** netmask (65,536 IP addresses) and **/28** netmask (16 IP addresses)
- ▶ After you define your VPC, you define your subnets, which are within the range of the VPC. For example, a VPC with 2 subnets:

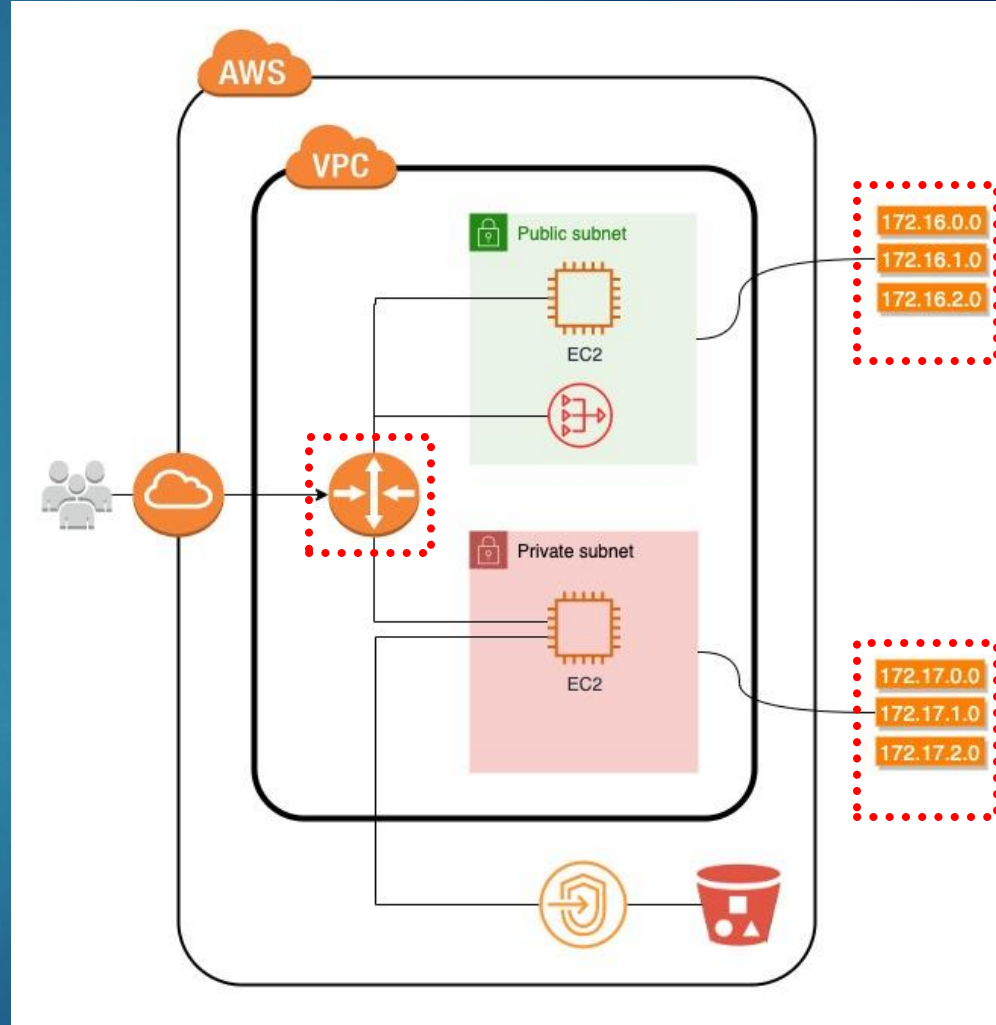
Network Component	CIDR	# of IP addresses	IP Range
VPC	10.0.0.0/24	256	10.0.0.0 – 10.0.0.255
Subnet #1	10.0.0.0/26	64	10.0.0.0 - 10.0.0.63
Subnet #2	10.0.0.64/26	64	10.0.0.64 - 10.0.0.127

Route Table

- ▶ A route table is a database that contains rules, or routes, that determine how traffic is directed within a Virtual Private Cloud (VPC)
- ▶ The rules specify which network traffic is directed to which network interface
- ▶ Each route in a table specifies a Destination and a Target
- ▶ For example, to enable your subnet to access the internet through an internet gateway, add the following route to your subnet route table:

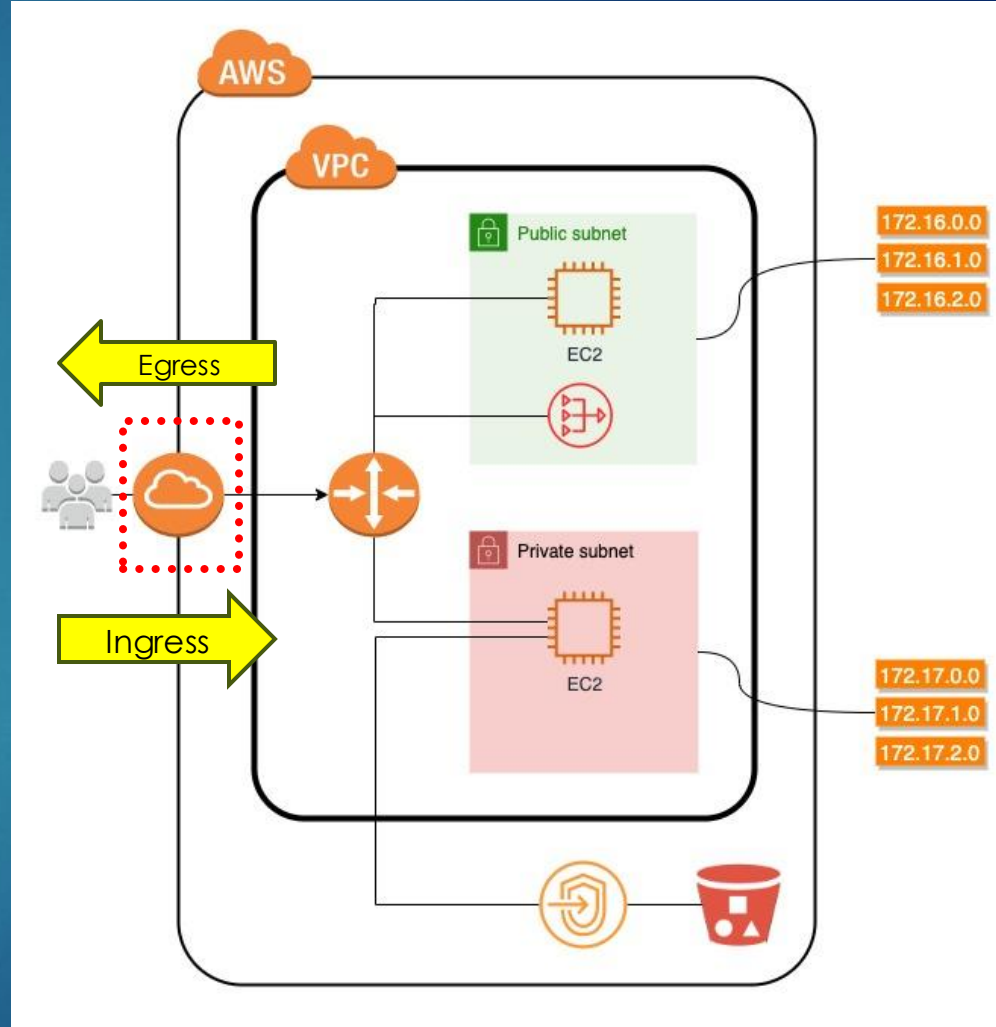
Destination	Target
0.0.0.0/0	igw-12345678901234567

- ▶ The **destination** for the route is 0.0.0.0/0, which represents all IPv4 addresses
- ▶ The **target** is the internet gateway that's attached to your VPC



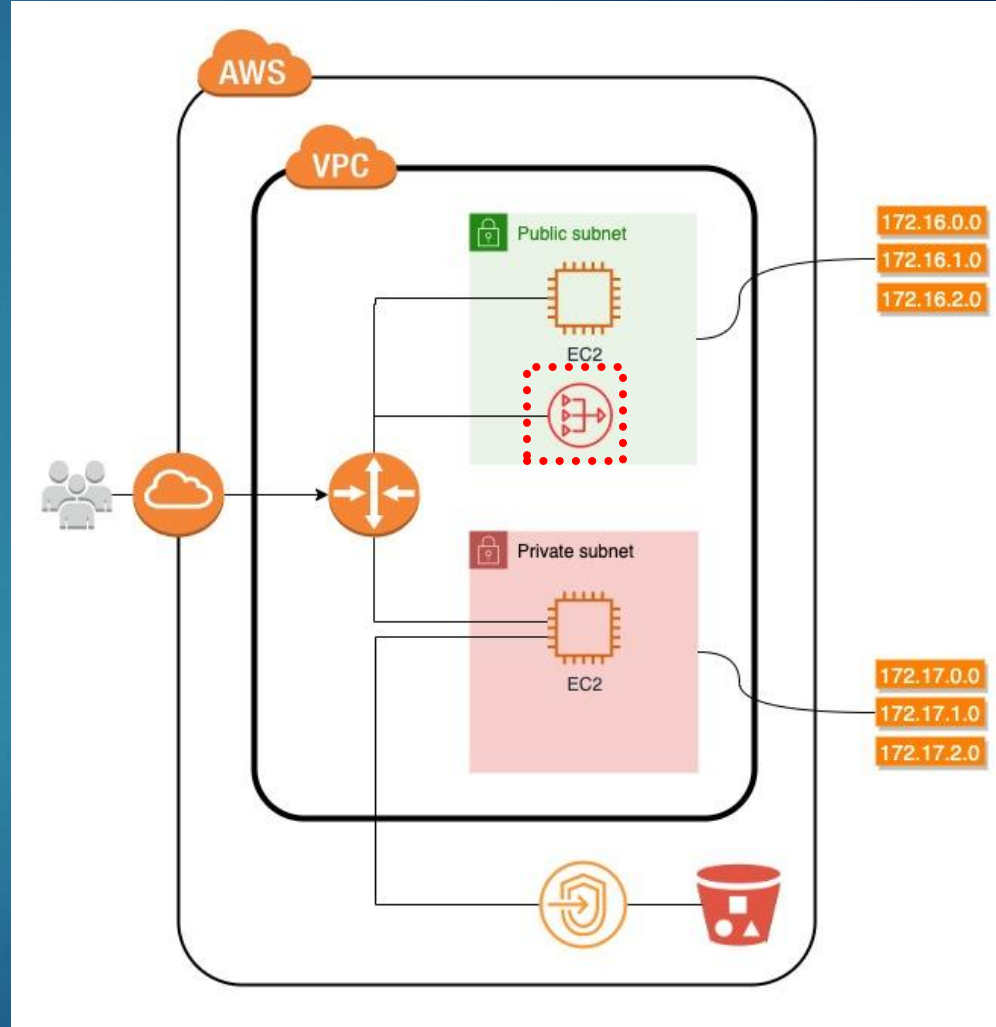
Internet Gateway

- ▶ An Internet Gateway (IGW) allows communication between instances in a VPC and the internet
- ▶ It acts as a “bridge” between a VPC and the internet, allowing you to route internet traffic to your instances
- ▶ Each VPC has only one IGW
- ▶ Ingress
 - ▶ Refers to data entering a system or network
- ▶ Egress
 - ▶ Refers to data leaving a system or network



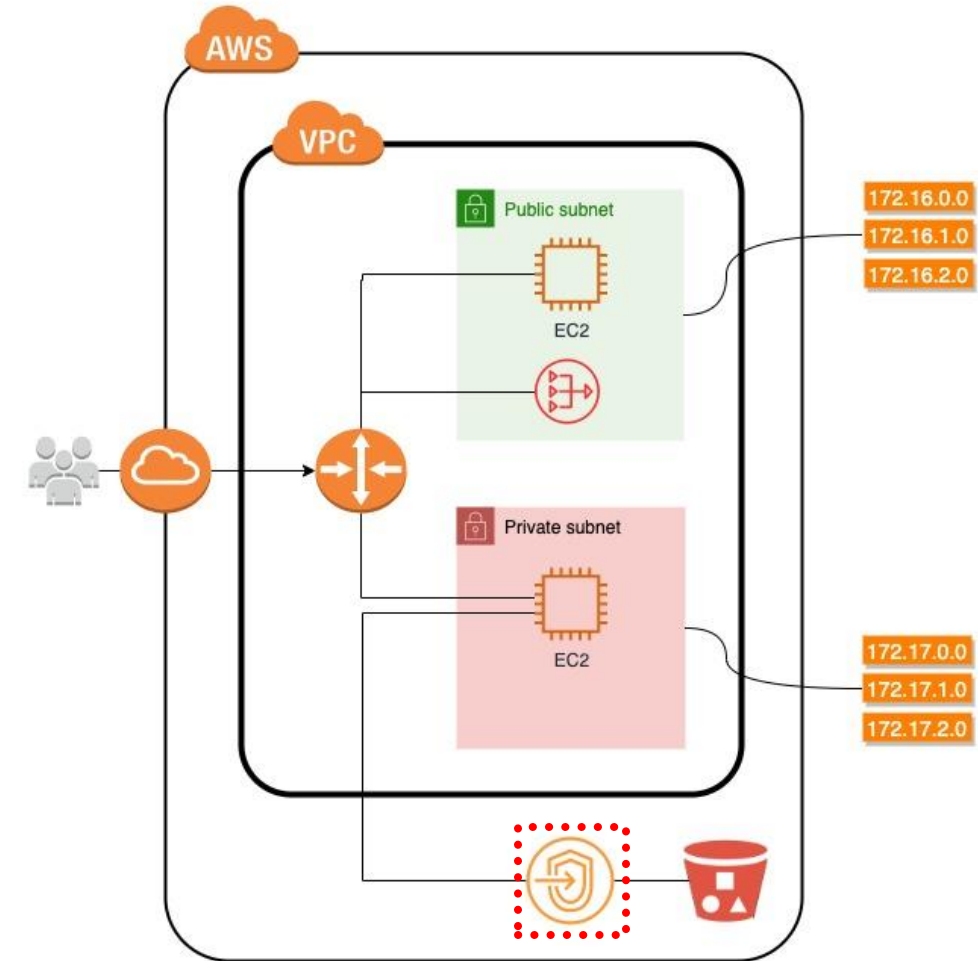
Network Address Translation (NAT) Gateway

- ▶ A NAT Gateway is used to enable instances present in a private subnet to help connect to the internet or AWS services
- ▶ It also makes sure that the internet doesn't initiate a connection with these instances
- ▶ To create a NAT gateway, you must specify the public subnet in which the NAT gateway should reside
- ▶ You must also specify an Elastic IP address to associate with the NAT gateway when you create it

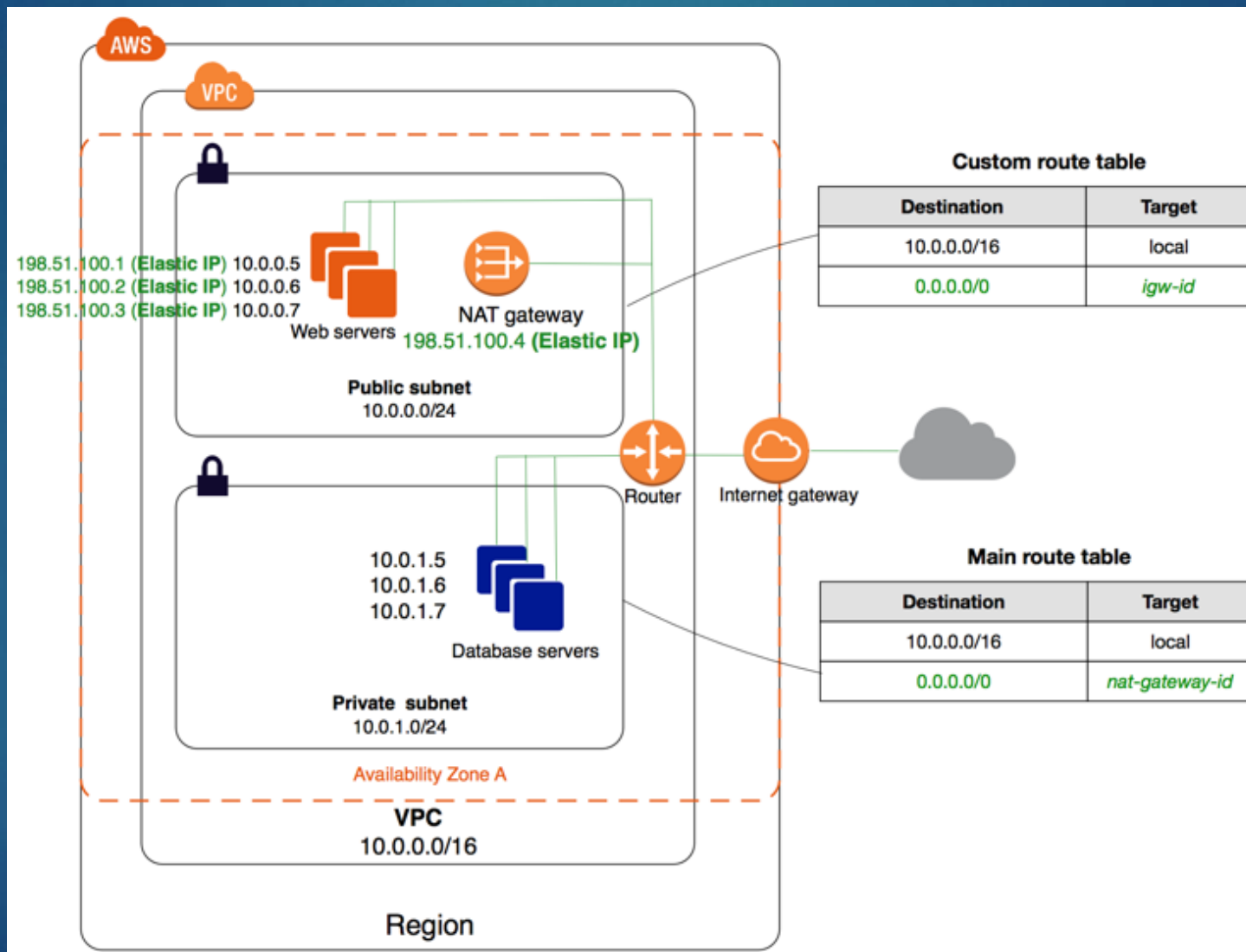


VPC Endpoint

- ▶ A VPC endpoint enables private connections between your VPC and supported AWS services (e.g. S3)
- ▶ They allow traffic to flow between a VPC and other services without ever leaving the Amazon network
- ▶ This introduces several benefits, not the least of which is improved network security

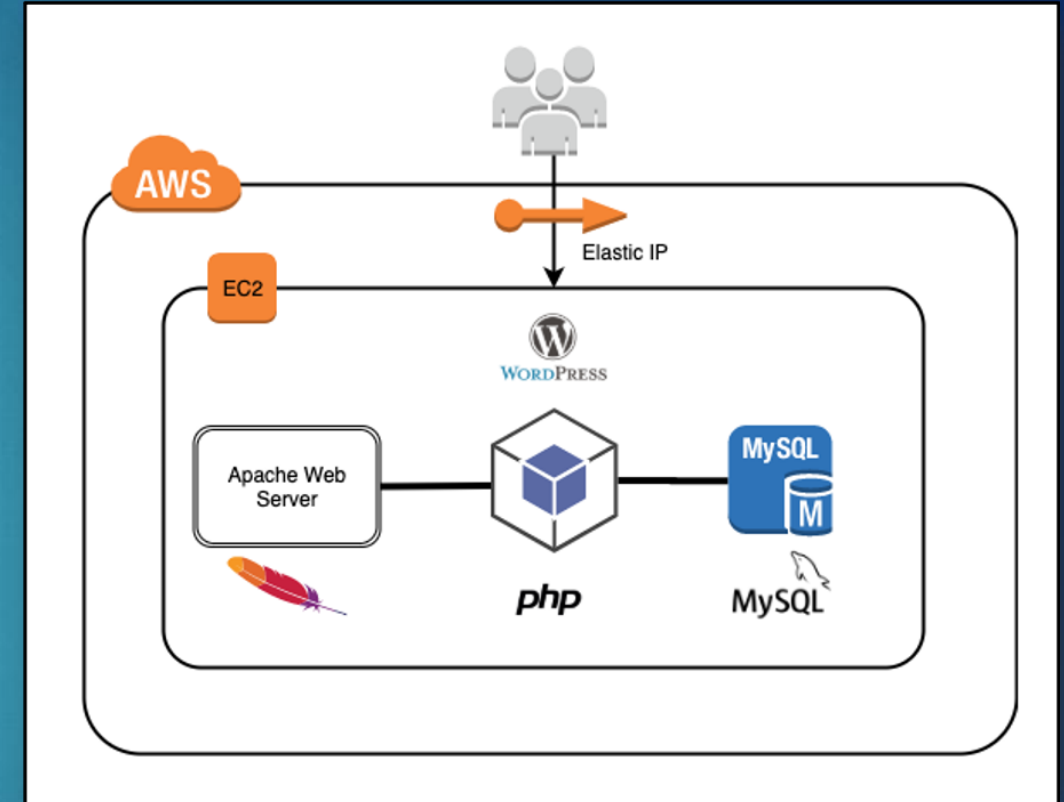


Tying this all together...



VPC and Subnets

- ▶ For the WordPress EC2 instance you created, was it deployed in a VPC?
- ▶ For the WordPress EC2 instance you created, what subnet was it deployed to:
 - ▶ Public
 - ▶ Private
 - ▶ We haven't created it yet



▼ Network settings [Get guidance](#)

VPC - required [Info](#)

vpc-055d45f9bdb0fb33c (default) ▼
172.31.0.0/16

Subnet [Info](#)

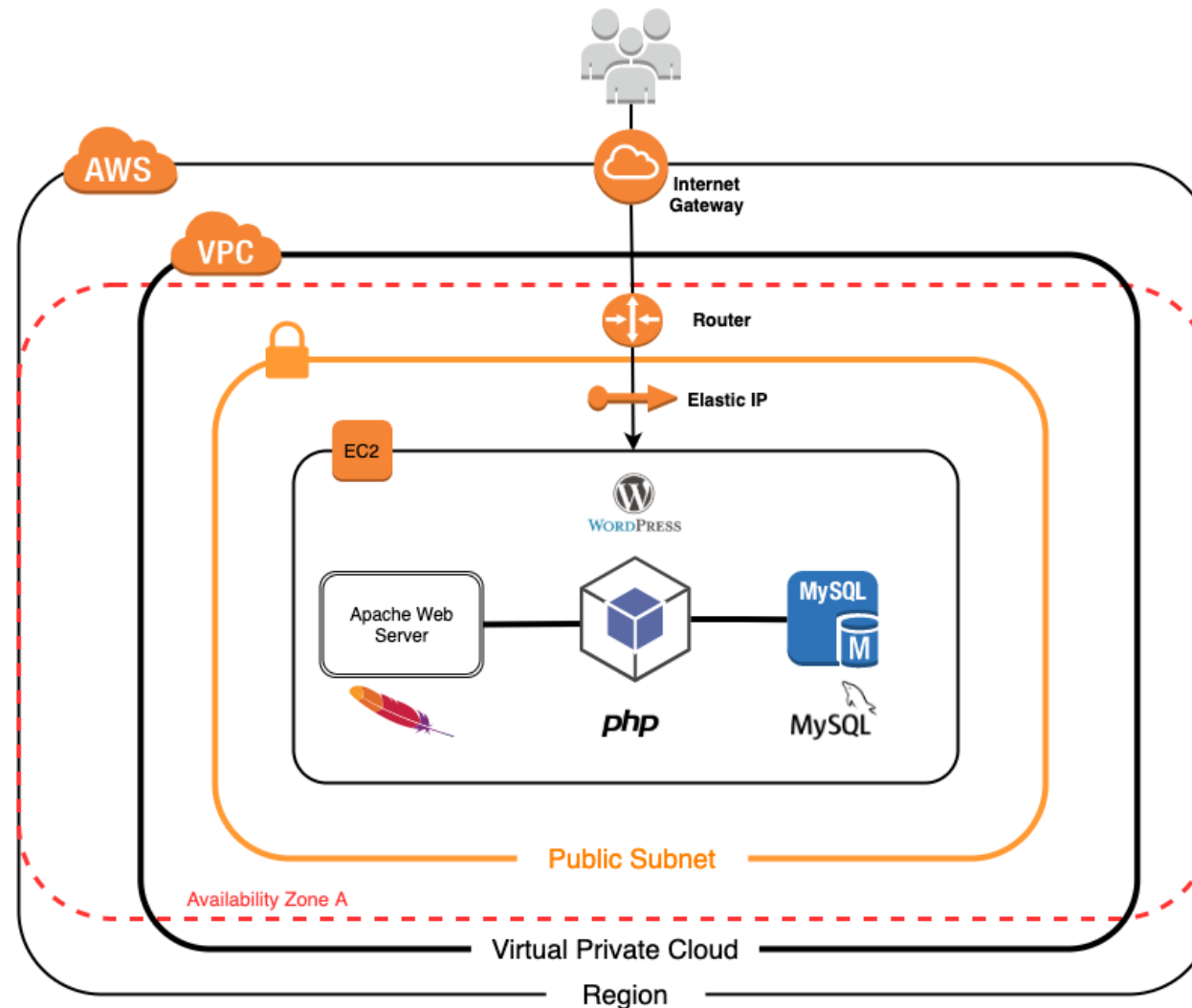
No preference ▼

↻ Create

All EC2 instances are launched in your default VPC and public subnet, which comes with your account

VPC and Subnets

37



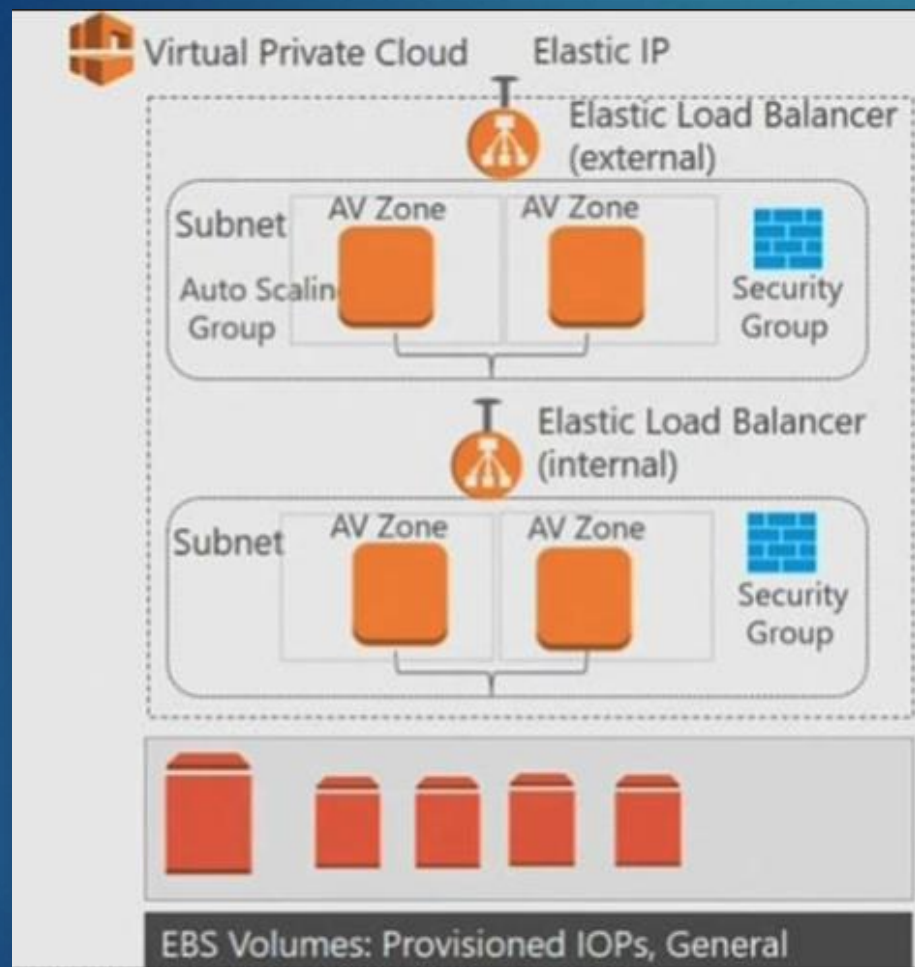
A VPC is located in a Region

Within a Region there are 2 or more Availability Zones

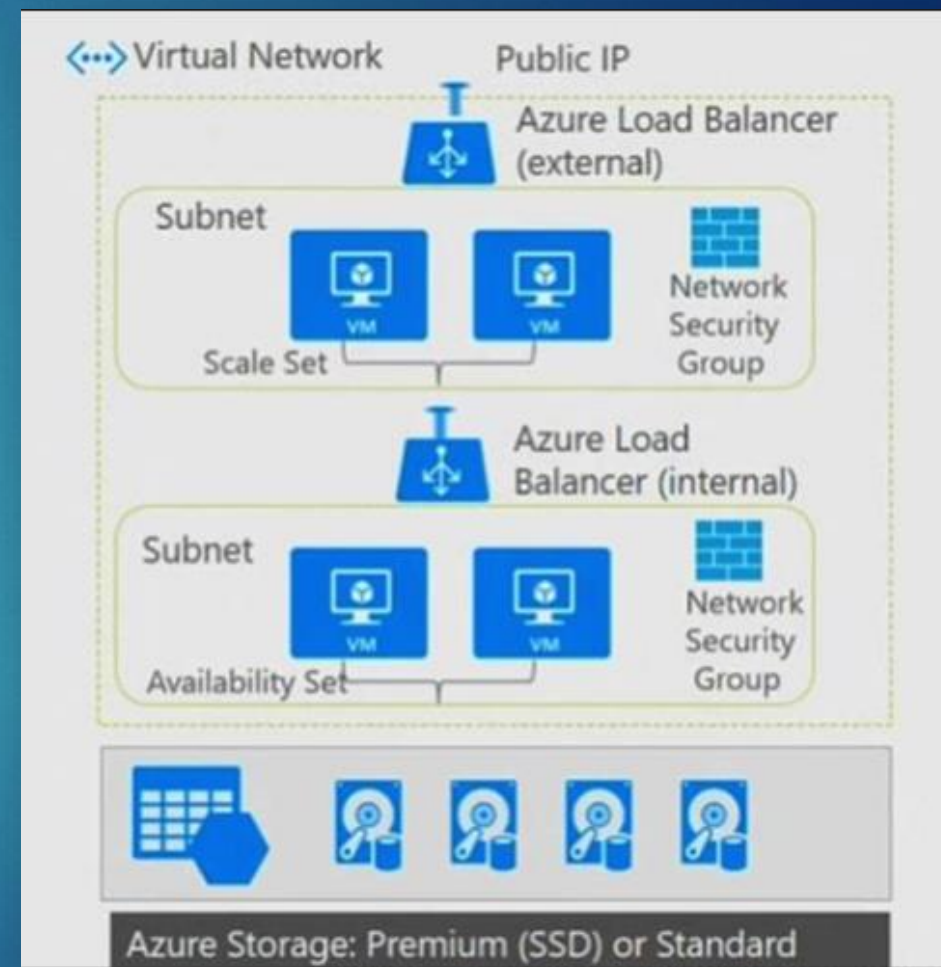
A Subnet is assigned to an Availability Zone

EC2

VPC and Subnets – AWS vs. Azure



AWS



Azure

Class Activity

- ▶ Go to myCourses to **Assignments > Activity – Creating a VPC and Subnets**
 - ▶ Remember to take a screenshot when you are done

